SIMATS ENGINEERING

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Saveetha Institute of Medical And Technical Sciences
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSA16 Data Warehousing and Data Mining

Assignment 2 – CO2

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CO2 Coverage: *Data mining from a database perspective, data mining functionalities, steps in the data mining process (KDD), architecture and classification of data mining systems, basic data mining tasks and issues, data dispersion, data cleaning, integration, transformation, reduction and discretization.*

Course Outcome: CO2 – Analyse data dispersion techniques and list the various algorithms available for mining the data.

Nature: Individual | **Duration:** 1 Week | **Total Marks:** 50

Q1 – Matching Business Problems to Mining Functionalities — 5 Marks

Problem Context – Telecom Subscriber Analytics

TeleNova is a telecom operator with 8 million subscribers. Its analytics team has received the following six business problems:

- **P1:** Find which plans and add-on services customers typically purchase together.
- **P2:** Estimate whether a given customer will leave the network next month.
- **P3:** Divide subscribers into natural usage segments, without any predefined labels.
- **P4:** Identify SIM cards whose calling behaviour is very different from all others.
- **P5:** Summarise the general characteristics of high-value customers and show how they differ from low-value customers.
- **P6:** Study how monthly data usage of customers has changed over the last 24 months.

Your Challenge

For each problem, identify:

- The **data mining functionality** involved (concept/class description, association, classification, prediction, clustering, outlier analysis or evolution analysis)
- Whether it is **supervised or unsupervised**, and the kind of pattern or knowledge produced
- The data attributes required and the data source (database or warehouse)
- At least two **candidate algorithms** from the syllabus and the reason for choosing them

Design Decision

The marketing manager proposes buying a single “all-purpose mining algorithm” for all six problems. Evaluate this proposal using your analysis.

Evidence Expected

A table of six problems mapped to functionality, learning type, data and algorithms, and a reasoned evaluation of the single-algorithm proposal.

Q2 – The KDD Process, System Architecture and Mining Issues — 5 Marks

Problem Context – Hospital Readmission Mining

MediCore, a hospital network, wants to discover factors that cause patients to be readmitted within 30 days. Patient data is spread across admission, laboratory, pharmacy and billing systems.

Your Challenge

- Map each KDD step (selection, preprocessing, transformation, mining, interpretation/evaluation) to **specific activities** MediCore must perform
- Draw the architecture of a typical data mining system for MediCore (database/warehouse server, knowledge base, mining engine, pattern-evaluation module, user interface) and mark where each requirement is handled
- Classify the system on four views: kinds of data, kinds of knowledge, techniques used and applications
- Identify **at least five data mining issues** specific to this scenario (methodology, user interaction, efficiency and scalability, diversity of data types, privacy/social) and a mitigation for each

Inquiry

A hospital director says: “We already have SQL and statistics, so we do not need data mining.” Using an example from MediCore, explain how a mining task differs from a database query and from a basic statistical summary.

Evidence Expected

KDD mapping, architecture diagram, system classification, issue–mitigation table and a reasoned answer to the director’s statement.

Q3 – Describing and Comparing Data Dispersion **7 Marks**

Problem Context – E-Commerce Delivery Times

An e-commerce company recorded the delivery time (in minutes) of 24 orders from one dark store:

28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43, 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95

Your Challenge

- Compute the **mean, median, mode, midrange**, and comment on what the relationship between mean and median suggests about the distribution
- Compute the **range, quartiles Q1 and Q3, IQR, variance and standard deviation**, showing your method (state which quartile convention you use)
- Prepare the **five-number summary** and draw a **boxplot**; apply the **1.5 × IQR rule** to identify outliers
- Remove the largest value and recompute all measures. Which measures changed the most, and why?
- Classify mean, median, variance, standard deviation and range as **distributive, algebraic or holistic** measures, and explain what this means when the data is split across many warehouse partitions
- Verify your manual results using a tool (spreadsheet, Python, Weka or KNIME statistics node) and explain any difference

Inquiry

The operations team wants to promise customers a “typical” delivery time. Which measure would you recommend and why? What would you report in addition, and why?

Evidence Expected

Manual calculations, boxplot, outlier identification, measure classification, tool verification and a reasoned recommendation.

Q4 – Cleaning Noisy and Incomplete Data **6 Marks**

Problem Context – Smart-Meter Energy Readings

A utility collects hourly readings from household smart meters. A sample extract is shown below:

Meter	Household type	kWh	Voltage (V)	Region
M101	Apartment	5	230	North

M102	Apartment	(missing)	231	North
M103	House	38	229	north
M104	House	-14	232	South
M105	(missing)	19	2300	South
M106	Apartment	11	230	S.

The sorted kWh values of 12 other hourly readings are: **5, 9, 11, 14, 18, 19, 23, 26, 27, 30, 33, 38**

Your Challenge

- Identify **every defect** in the extract and classify it (missing value, noise, inconsistency, invalid value)
- For each defect, propose and justify a handling strategy from: ignore the tuple, fill manually, global constant, attribute mean, class-wise mean, or most probable value (regression/inference)
- Apply **equal-frequency binning (depth 4)** to the 12 sorted values and smooth by **bin means** and by **bin boundaries**, showing all bins
- Explain how regression and clustering could be used to detect or smooth noise, and why a reading such as 2300 V must be treated differently from a reading of 5 kWh

Inquiry

Replacing every missing value with the global mean is quick. Explain **how this affects the dispersion of the attribute and the quality of patterns mined later**, and when it should be avoided.

Evidence Expected

Defect classification, strategy table, binning calculations and a reasoned discussion on the impact of imputation on dispersion and mining.

Q5 – Integration and Transformation of Data **7 Marks**

Problem Context – University Data Consolidation

A university combines its Admissions database (attributes: *student_id, name, dob, hostel_status, scholarship*) with its Learning Management System (attributes: *sid, full_name, attendance_pct, marks, scholarship_flag*).

Your Challenge

Part A – Integration

- Identify the entity-identification, naming, unit/format and value conflicts and show how they are resolved in the integrated schema
- Explain why integration often creates **redundant attributes**, and how correlation analysis helps detect them

Part B – Redundancy analysis

- For attendance % = {60, 70, 75, 80, 85, 90} and marks = {52, 58, 66, 70, 79, 85}, compute the **Pearson correlation coefficient** and interpret it
- For the observed counts below, compute the **χ^2 statistic** (with expected values) and state whether *hostel status* and *scholarship* are independent at the 0.05 level (critical value for 1 degree of freedom = 3.84)

	Scholarship	**No scholarship**	**Total**
Hostel	120	180	300
Day scholar	80	320	400
Total	200	500	700

Part C – Transformation

- Annual family income (₹) of five students: **20,000; 35,000; 50,000; 80,000; 150,000**. Normalise using **min-max (0 to 1), z-score and decimal scaling**

- Two students have (age, income) = (19, 35,000) and (21, 80,000). Compute their Euclidean distance **before and after** min-max normalisation of both attributes (assume age range 17–25 and income range 20,000–150,000) and explain what this reveals for distance-based mining
- Give one example each of **aggregation, generalisation and attribute construction** for this data

Evidence Expected
Conflict-resolution table, correlation and χ^2 calculations with interpretation, three normalisation methods and a before/after distance comparison.

Q6 – Data Reduction and Discretization — 7 Marks

Problem Context – Industrial Sensor Monitoring
A factory logs **200 sensor attributes** from machines, producing **5 million rows** per month. Analysts need faster mining without losing the ability to detect machine failure.

Your Challenge
Part A – Choosing reduction strategies

- For each category, describe **one technique and when it is suitable** for this data: data-cube aggregation, attribute subset selection (forward selection, backward elimination, decision-tree induction), dimensionality reduction (PCA), numerosity reduction (sampling, histograms, clustering) and compression
- Compare **simple random sampling without replacement, with replacement, cluster sampling and stratified sampling** and state which is appropriate if only 0.5% of the rows are failures
- Explain how many principal components you would keep and the criterion you would use

Part B – Discretization by calculation
Temperature readings: 3, 7, 7, 9, 12, 14, 14, 15, 18, 21, 21, 22, 25, 25, 27, 31, 36, 36, 40, 44

- Discretize into 4 intervals using **equal-width** and **equal-frequency** binning. Show the intervals and the number of values in each
- Using the labelled values below, find the best **entropy-based split point** (candidate split points are midpoints between adjacent values; show the information gain of each candidate):

Value	10	12	15	18	22	25	28	31	35	40
Class	N	N	N	N	Y	Y	N	Y	Y	Y

- Compare the unsupervised and supervised discretization results and explain which suits a classification task.

Evidence Expected
Reduction strategy justification, sampling comparison, equal-width/equal-frequency intervals, information-gain table and a reasoned comparison.

DAY-4 DESIGN CHALLENGE

Q7 – Diagnose, Redesign and Validate — 13 Marks

Problem Context – Loan-Default Pipeline Failure
A retail bank built a model to predict loan default. Its preprocessing pipeline was:

1. Missing *income* values were replaced by the global mean income.
2. The attribute *occupation* (18% missing) was dropped.
3. *Income* (highly right-skewed) was discretized into 5 **equal-width** intervals.
4. All numeric attributes were min-max normalised using the **complete dataset**, before splitting into training and test sets.
5. A random 10% sample of rows was used for training. Only 3% of loans in the data are defaults.

The model reported **97% accuracy** on the test set. After deployment, it missed most defaulters and the bank lost money.

A. Diagnose the Computational Bottleneck – 2 Marks

Identify the defects in the pipeline. For each, name the **data preprocessing principle violated** and explain how it can mislead the model or the evaluation.

B. Define the Design Objective – 2 Marks

State measurable objectives (for example the metric to improve and the acceptable trade-off) and the constraints that the redesigned pipeline must satisfy.

C. Design the Revised Preprocessing Strategy – 4 Marks

Design an **ordered** pipeline that addresses each defect using appropriate cleaning, integration, transformation, reduction and discretization techniques. For each step, justify the technique chosen over at least one alternative and state whether it is computed on training data only.

D. Predict Before Implementing – 1 Mark

Before running anything, predict how your redesign will change: the dispersion of *income*, the class distribution in the training data, and accuracy versus recall on defaulters.

E. Implement and Validate the Redesign – 2 Marks

Implement the original and the redesigned pipelines on a suitable dataset (for example UCI *German Credit* or a similar public credit dataset) using Weka, KNIME or Python. Compare them with the same test data and report suitable metrics.

F. Defend the Final Design – 2 Marks

Compare your predictions with the results. Defend your final pipeline, and state one risk that remains.

Evidence Expected

A diagnosis, measurable objectives, a justified pipeline, a written prediction, validated comparison results and a defence of the final design.

MARK DISTRIBUTION

Question	Assessed Area	Marks
Q1	Mining Functionalities and Algorithm Mapping	**5**
Q2	KDD Process, Architecture and Issues	**5**
Q3	Data Dispersion	**7**
Q4	Data Cleaning	**6**
Q5	Data Integration and Transformation	**7**
Q6	Data Reduction and Discretization	**7**
Q7	**Day-4 Design Challenge – Diagnose, Redesign and Validate**	**13**
TOTAL		**50**

Submission Requirements

Submit:

- Your assignment report containing the required evidence for Q1–Q7
- Spreadsheets/code/workflows used for calculations and validation
- Screenshots of tool outputs (Weka/KNIME/Python) with the settings used
- Datasets (or links) used in Q7

Manual calculations are **primary evidence**. Tool output may be used to verify your work but must not replace the working shown.

Assessment Verification

Students may be asked to explain any part of their:

- Manual calculations and their method
- Choice of preprocessing techniques
- Predictions and results
- Design Challenge

during a brief verification discussion/viva. The student should therefore be able to **explain and defend the evidence submitted as their own work**.

Overall Assessment Expectation

This assignment does not primarily assess your ability to reproduce textbook definitions or tool outputs.

It assesses your ability to:
Understand → Describe → Clean → Transform → Reduce → Predict → Validate → Defend

Your conclusions and design decisions must be supported by **your own reasoning and evidence**.

Q1 – Matching Business Problems to Mining Functionalities

TeleNova has six different analytical requirements. Each requirement corresponds to a different data-mining functionality because the type of knowledge requested is different.

Problem	Functionality	Learning / knowledge	Required data & source	Two suitable candidates + reason
P1	Association	Unsupervised; frequent co-occurrence rules such as Plan A → Add-on B.	Plan, add-on, customer/transaction ID, purchase date; telecom billing/CRM warehouse.	Apriori – discovers frequent itemsets and association rules; FP-Growth – efficiently mines frequent patterns.
P2	Classification / Prediction	Supervised; predicts whether a known customer will churn.	Customer tenure, plan, usage, complaints, payments, churn label; CRM/warehouse.	Decision Tree – interpretable churn rules; Naïve Bayes – simple probabilistic baseline for classification.
P3	Clustering	Unsupervised; produces natural subscriber groups without labels.	Usage, call duration, recharge, plan, data consumption; customer warehouse.	K-Means – efficient for large numeric datasets; Hierarchical clustering – shows nested customer groups.
P4	Outlier analysis	Usually unsupervised; identifies unusual calling behaviour.	Call frequency, duration, destinations, time-of-day, location; call-detail records.	Distance-based outlier detection – identifies observations far from peers; clustering-based detection – flags points weakly belonging to clusters.
P5	Concept/class description	Descriptive; summarises characteristics of a target class and contrasts it with another class.	Customer value, revenue, usage, plan, tenure, complaints; CRM/warehouse.	Decision Tree – produces understandable class descriptions; OLAP/generalisation – summarises attributes across customer classes.
P6	Evolution analysis	Descriptive/time-oriented; discovers trends and changes over time.	Monthly data usage, customer ID/segment, month, plan; historical warehouse.	Time-series/trend analysis – detects temporal movement; regression – quantifies systematic change over time.

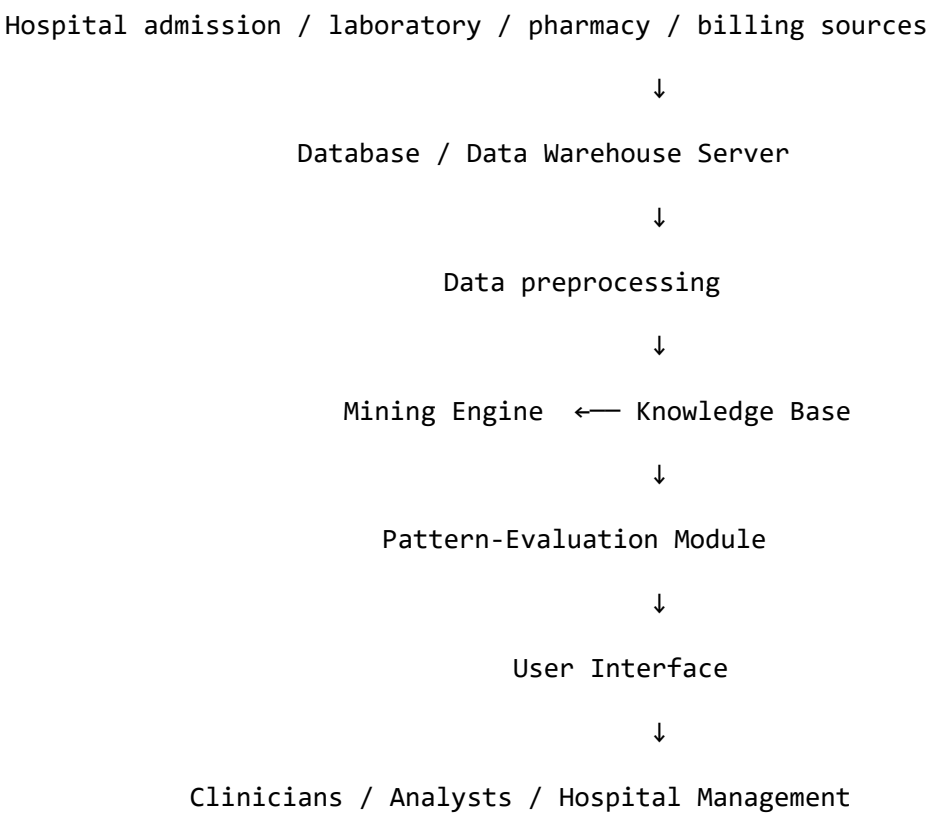
Evaluation of the single all-purpose algorithm: The proposal is not suitable. The six tasks have different objectives and learning settings. Association mining searches for co-occurrence rules, classification predicts a labelled outcome, clustering discovers groups without labels, outlier analysis searches for abnormal observations, concept description summarises classes, and evolution analysis studies change over time. A single algorithm may be technically adaptable, but it would not be equally appropriate, interpretable, efficient, or accurate for all six tasks. Therefore, TeleNova should select algorithms according to the mining functionality, data characteristics, scalability requirement, and business objective.

Q2 – The KDD Process, System Architecture and Mining Issues

MediCore must combine information from admission, laboratory, pharmacy and billing systems before mining factors related to 30-day readmission.

KDD step	MediCore activity	Output
1. Selection	Select patients, admission episodes and variables relevant to 30-day readmission from the four source systems.	Integrated analysis dataset.
2. Preprocessing	Handle missing laboratory values, duplicate patient records, inconsistent codes, impossible values and noisy measurements.	Clean dataset.
3. Transformation	Convert dates to lengths of stay, create readmission-within-30-days label, aggregate medication/lab information and normalise or discretise where appropriate.	Mining-ready attributes.
4. Data mining	Apply classification, association, clustering or outlier analysis to discover factors associated with readmission.	Rules, patterns, clusters or predictive models.
5. Interpretation / evaluation	Evaluate interestingness, accuracy and usefulness; remove weak or clinically meaningless patterns; present actionable findings.	Validated knowledge for hospital decisions.

Typical data-mining system architecture:



System classification:

View	Classification for MediCore	Explanation
Kinds of data	Relational, warehouse, temporal and possibly heterogeneous clinical data.	Patient, laboratory, pharmacy and billing records have different structures and time attributes.
Kinds of knowledge	Characterisation, classification/prediction, association and outlier knowledge.	The system can describe readmitted patients, predict readmission, find co-occurring risk factors and identify unusual records.
Techniques used	Machine learning, statistical analysis, database/OLAP methods and pattern discovery.	Different techniques are combined according to the mining task.

Applications	Healthcare analytics, risk prediction, quality improvement and resource planning.	The final knowledge supports interventions for high-risk patients.
Issue	**Why it matters**	**Mitigation**
Methodology / task selection	A method suitable for one clinical task may not suit another.	Define the mining objective first and select task-specific algorithms.
User interaction	Clinicians need understandable patterns, not only technical model output.	Use interpretable rules, visualisation and clinician-in-the-loop evaluation.
Efficiency and scalability	Large longitudinal records and laboratory data can be expensive to process.	Use indexing, aggregation, sampling where valid, parallel processing and efficient mining algorithms.
Diversity of data types	Numeric lab values, categorical diagnoses, prescriptions and dates require different treatment.	Use schema integration, type-aware preprocessing and suitable transformations.
Privacy / social concerns	Patient data contains highly sensitive information.	Apply access control, de-identification, secure storage, auditing and minimum-necessary data use.

SQL versus statistics versus data mining: An SQL query answers a specified question, such as "How many patients were readmitted within 30 days?" A statistical summary can calculate the mean length of stay or the percentage of readmitted patients. Data mining goes further by searching for previously unknown, potentially useful patterns, such as a combination of long stay, repeated abnormal laboratory results and medication-related factors that frequently occurs among readmitted patients. Thus, SQL retrieves known information, basic statistics summarise selected variables, while data mining systematically discovers patterns or predictive relationships.

Q3 – Describing and Comparing Data Dispersion

Given sorted delivery times (minutes): 28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43, 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95.

Mean = $\Sigma x/n = 1143/24 = 47.625$ minutes.

Median: There are 24 observations, so the median is the average of the 12th and 13th observations = (43 + 45)/2 = 44 minutes.

Mode: 35 and 42 both occur twice; therefore the data is bimodal.

Midrange = (minimum + maximum)/2 = (28 + 95)/2 = 61.5 minutes.

Since the mean (47.625) is greater than the median (44), the distribution is positively/right skewed. The high value 95 pulls the mean upward.

Quartile convention used: median-of-halves (Tukey-style) convention.

Lower half = 28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43 → Q1 = (35 + 36)/2 = 35.5.

Upper half = 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95 → Q3 = (53 + 55)/2 = 54.0.

Therefore IQR = Q3 − Q1 = 54 − 35.5 = 18.5 minutes.

Range = 95 − 28 = 67 minutes. Variance (population form) = 213.318 min^2 and standard deviation = 14.605 minutes. The sample variance would be 222.592 min^2 with sample SD 14.920.

Five-number summary = Minimum 28, Q1 35.5, Median 44, Q3 54, Maximum 95.

1.5×IQR fences: Lower fence = 35.5 − 1.5(18.5) = 9.25; Upper fence = 54 + 1.5(18.5) = 81.75. Since 95 > 81.75, 95 minutes is an outlier.



After removing the outlier 95:

n = 23; mean = 45.565 minutes; median = 43.0 minutes; mode remains 35 and 42; midrange = (28 + 72)/2 = 50.0 minutes; range = 72 − 28 = 44 minutes.

Using the same median-of-halves convention for 23 observations: the overall median is 43. The lower 11 observations have median Q1 = 36, and the upper 11 observations have median Q3 = 53. Therefore IQR = 17 minutes.

Population variance after removal = 120.767 min² and population standard deviation = 10.989 minutes.

The measures most affected by removing 95 are the mean, range, midrange, variance and standard deviation because these measures use the magnitude of extreme observations. The median and mode are much more resistant to the single extreme value.

Measure	Type	Meaning for partitioned data
Mean	Algebraic	Can be computed from partition-level count and sum, then combined.
Median	Holistic	Generally requires information about the ordered distribution and cannot be obtained exactly from a fixed-size summary of every partition.
Variance	Algebraic	Can be combined from partition count, sum and sum of squares.
Standard deviation	Algebraic	Derived from variance and therefore can be combined using algebraic summaries.
Range	Distributive	Global minimum and maximum can be obtained from partition minima and maxima.

Recommendation: Use the median (43–44 minutes depending on whether the outlier is retained) as the typical delivery time because it is resistant to the extreme 95-minute delay. Report the IQR as well because it describes the spread of the central 50% of orders. The business could also separately report the upper-tail delay/outlier rate.

Q4 – Cleaning Noisy and Incomplete Data

The extract contains missing values, inconsistent values, noise and invalid values.

Record / field	Defect	Classification	Suitable handling
M102 household type	Missing	Missing value	Use class-wise/most-probable inference if enough related records exist; manual fill is

			preferable when the household type is known.
M103 household type = House; direction = north	north vs North	Inconsistency	Standardise categorical values to a single representation such as North.
M104 kWh = −14	Negative energy consumption	Invalid value / possible noise	Verify source meter. If impossible physically, mark as missing or invalid rather than treating −14 as genuine consumption.
M105 household type	Missing	Missing value	Infer using other household attributes or class-wise/most-probable value; manual correction if source record is available.
M105 voltage = 2300 V	Extreme / implausible for stated household context	Invalid value / noise	Check decimal/unit/transcription error; do not blindly smooth it as ordinary noise.
M106 direction = S	S vs South	Inconsistency	Map abbreviations to standard categorical values, e.g. S → South.

Handling strategy: Ignoring a tuple is appropriate only when the record is unusable and missingness is small. Manual filling is best when the correct value can be verified from the source. A global constant is simple but can create an artificial mode. Attribute mean is useful for approximately symmetric numeric data but can reduce variance. Class-wise mean is preferable when the attribute differs systematically between groups. Most-probable-value methods use other attributes and relationships to infer a plausible value.

Equal-frequency binning with depth 4 for the sorted values 5, 9, 11, 14, 18, 19, 23, 26, 27, 30, 33, 38:

Bin	Values	Bin mean	Mean smoothing	Boundary smoothing
B1	5, 9, 11	8.33	8.33, 8.33, 8.33	5, 11, 11
B2	14, 18, 19	17.00	17, 17, 17	14, 19, 19
B3	23, 26, 27	25.33	25.33, 25.33, 25.33	23, 27, 27
B4	30, 33, 38	33.67	33.67, 33.67, 33.67	30, 30/38, 38

For boundary smoothing, each value is replaced by the nearer bin boundary. For the middle value 33 in B4, both boundaries 30 and 38 are equally distant, so either boundary can be used if the tie rule is stated consistently.

Regression-based noise handling: fit a model using related attributes such as temperature, time, household type and neighbouring readings; a large residual may indicate an anomalous reading. Clustering-based handling: group similar records and identify observations far from their cluster centre as potential noise.

A 2300 V value must be treated differently from 5 kWh because they are different attributes with different units and physical meanings. A value may be numerically large without being comparable across attributes. The first question is whether 2300 V is physically plausible, a unit/decimal error, or a sensor fault.

Global-mean imputation reduces dispersion because many missing observations are replaced by the same central value. It can reduce variance, shrink correlations, create artificial concentration and change clustering, classification and association patterns. It should be avoided when missingness is informative, when the distribution is highly skewed, or when class/group differences are important.

Q5 – Integration and Transformation of Data

The Admissions and LMS systems describe the same student entity using different names and structures. Integration requires entity matching, schema alignment and value standardisation.

Conflict	Example	Resolution
Entity identification	Admissions student_id vs LMS sid	Use a common student identifier and maintain a cross-reference table if legacy IDs differ.
Naming conflict	name vs full_name	Map both to a standard attribute such as student_name.
Format conflict	dob may be stored in different date formats	Convert to one standard date format such as YYYY-MM-DD.
Unit / scale conflict	attendance_pct must be represented consistently as percentage, not mixed fraction/percentage.	Standardise to 0–100 percentage.
Value conflict	hostel_status and scholarship_flag may use different codes such as Y/N, Yes/No, 1/0.	Create controlled domains, e.g. Yes/No, and document the mapping.
Redundant attribute	scholarship and scholarship_flag may represent the same concept.	Keep one authoritative field after validation.

Integration often creates redundancy because the same real-world fact appears in multiple systems under different names or at different granularities. Redundancy can increase storage and may create inconsistent information. Correlation analysis can identify numerical attributes carrying nearly the same information, while domain knowledge and schema analysis are needed for categorical redundancy.

Pearson correlation for attendance A = [60.0, 70.0, 75.0, 80.0, 85.0, 90.0] and marks M = <built-in method tolist of numpy.ndarray object at 0x7f3aa48466e0>:

Formula: $r = [n\Sigma xy - (\Sigma x)(\Sigma y)] / \sqrt{([n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2])}$.

Using the six paired observations gives $r \approx 0.9851$. This is a very strong positive correlation: students with higher attendance in this small sample tend to have higher marks. Correlation does not by itself prove that attendance causes marks.

Chi-square test for independence:

	Scholarship	No scholarship	Row total
Hostel	120	180	300
Day scholar	80	320	400
Column total	200	500	700
Cell	**Observed O**	**Expected E = row×column/700**	**$(O-E)^2/E$**
Hostel–Scholarship	120	85.71428571428571	13.714285714285719
Hostel–No scholarship	180	214.28571428571428	5.485714285714283
Day scholar–Scholarship	80	114.28571428571429	10.285714285714288
Day scholar–No scholarship	320	285.7142857142857	4.114285714285712

$\chi^2 = 33.600$. Since 33.600 > 3.84 at 0.05 significance and df = 1, reject the null hypothesis of independence. Therefore hostel status and scholarship are statistically associated in the observed table.

Part C – Normalisation of annual family income

Income (₹)	Min-max (0–1)	Z-score	Decimal scaling
20,000	0.0000	-1.0217	0.020
35,000	0.1154	-0.6957	0.035
50,000	0.2308	-0.3696	0.050
80,000	0.4615	0.2826	0.080
150,000	1.0000	1.8043	0.150

Min-max formula: $v' = (v - \min)/(\max - \min)$. Z-score formula: $v' = (v - \mu)/\sigma$, using population standard deviation for this calculation. Decimal scaling: $v' = v/10^j$, where j is the smallest integer that makes all transformed absolute values less than 1; here j = 6.

Euclidean distance before normalisation = $\sqrt{[(21-19)^2 + (80{,}000-35{,}000)^2]} \approx 45{,}000.00$.

After min-max normalisation using age range 17–25 and income range ₹20,000–₹150,000, the distance ≈ 0.4270.

This shows why transformation matters for distance-based mining. Before scaling, income dominates the distance because its numerical magnitude is much larger than age. After normalisation, age and income contribute on comparable scales.

Transformation	Example for university data
Aggregation	Convert daily LMS attendance records into monthly average attendance.
Generalisation	Replace city → district → state, or exact age → age group.
Attribute construction	Construct attendance-to-marks ratio or a binary high-risk indicator from existing attributes.

Q6 – Data Reduction and Discretization

The factory has 200 sensor attributes and 5 million rows per month. The objective is to reduce computational cost while retaining information needed for failure detection.

Technique	Application	Benefit / caution
Cube aggregation	Aggregate readings by machine, hour/day, shift or production line.	Reduces rows while retaining summaries; excessive aggregation may hide short failures.
Attribute subset selection	Use forward selection, backward elimination or decision-tree-based selection to retain useful sensors.	Reduces dimensionality; must preserve failure-related attributes.
PCA	Transform correlated sensor attributes into fewer principal components.	Captures major variance with fewer dimensions; components may be harder to interpret.
Numerosity reduction	Use sampling, histograms or clustering to represent many records compactly.	Reduces number of stored/processed records; rare failures must not be lost.
Compression	Compress stored sensor data using suitable lossless/lossy methods.	Reduces storage and transfer cost; lossy compression must not remove diagnostic signals.

Sampling comparison for a 0.5% failure rate:

Method	Description	Suitability for rare failures
Simple random sampling without replacement	Selects a fixed number of distinct records.	May contain very few or zero failures; risky if the sample is small.
Simple random sampling with replacement	Samples records independently and may select the same record more than once.	Can duplicate common normal records and is not ideal for rare-event coverage.
Cluster sampling	Selects groups such as machines or production batches.	Efficient operationally, but a selected cluster may contain mostly normal records and may miss rare failures.
Stratified sampling	Split records into strata such as failure/non-failure, then sample from each stratum.	Best choice here because the rare failure class can be deliberately represented while preserving known proportions or a controlled oversampling ratio.

PCA criterion: retain enough principal components to explain a chosen high proportion of total variance, for example 90–95%, while checking that failure-detection performance does not deteriorate. PCA should be fitted on training data only when used for predictive modelling.

Equal-width discretization for sensor temperature values 3, 7, 7, 9, 12, 14, 14, 15, 18, 21, 21, 22, 25, 25, 27, 31, 36, 36, 40, 44: minimum = 3, maximum = 44. With 4 bins, width = (44−3)/4 = 10.25. Approximate intervals are [3,13.25), [13.25,23.5), [23.5,33.75), [33.75,44].

Equal-width bin	Values
B1	3, 7, 7, 9, 12

B2	14, 14, 15, 18, 21, 21, 22
B3	25, 25, 27, 31
B4	36, 36, 40, 44

Equal-frequency discretization attempts to place approximately the same number of observations in each bin. With 20 values and 4 bins, a rank-based grouping gives five values per bin: B1 = 3,7,7,9,12; B2 = 14,14,15,18,21; B3 = 21,22,25,25,27; B4 = 31,36,36,40,44. Because 21 occurs at the boundary, the tie-breaking rule should be documented.

Entropy-based supervised discretization: The class labels for values 10,12,15,18,22,25,28,31,35,40 are N,N,N,N,Y,Y,N,Y,Y,Y. The initial entropy is 1.000 bit.

Candidate split	Information gain
11	0.1080
13.5	0.2365
16.5	0.3958
20	0.6100
23.5	0.2781
26.5	0.1245
29.5	0.3958
33	0.2365
37.5	0.1080

The highest information gain is approximately 0.6100 at split 20, so 20 is the best candidate among the tested midpoints. This split gives the strongest class separation according to entropy reduction.

Supervised discretization uses class labels to choose boundaries that improve class purity, so it is useful for predictive modelling. Unsupervised discretization does not use labels and instead uses distribution-based rules such as equal-width or equal-frequency binning. Supervised discretization may improve predictive usefulness, but it must be performed without leaking test-set labels into training.

Q7 – Diagnose, Redesign and Validate

The original loan-default pipeline produces 97% accuracy but still misses most defaulters. The central problem is that accuracy is misleading when defaults are only 3% of the population.

A. Diagnose the computational bottleneck / defects

Original step	Defect	Violated principle	Why it is misleading
1. Global mean income imputation	Uses one overall mean for all missing incomes.	Appropriate imputation / distribution preservation.	Can reduce variance, distort skewness and ignore relationships with occupation, age or other variables.
2. Drop occupation with 18% missing	Discards a potentially useful attribute.	Preserve useful information; handle missingness appropriately.	Loses predictive information and may introduce bias if missingness is not random.
3. Equal-width income bins	Income is highly right-skewed.	Use distribution-aware transformation/discretization.	Most observations may fall into a few bins while extreme-income regions receive wide, sparsely populated intervals.
4. Normalise before split	Uses complete-data statistics including future test records.	Prevent data leakage.	Test-set information influences training preprocessing and can make evaluation optimistic.
5. Random 10% training with 3% defaults	Training set is likely to contain very few defaults.	Class imbalance handling.	Model can learn to favour the majority class and still obtain high accuracy while missing defaults.
6. Evaluate mainly with accuracy	97% accuracy is treated as success.	Use metrics aligned with the business objective.	A model can obtain near-majority accuracy while having poor recall for the costly

			minority class.

B. Design objective and constraints

Primary measurable objective: improve recall/sensitivity for actual defaulters while maintaining acceptable precision and overall operational workload. Secondary measures should include precision, F1-score, specificity, balanced accuracy and the confusion matrix; ROC-AUC or PR-AUC can be reported when appropriate.

Constraints: avoid test-data leakage; preserve a final untouched test set; handle missing values without unnecessarily deleting useful attributes; retain the rare default class; keep preprocessing reproducible; and choose thresholds according to the bank's relative costs of false negatives and false positives.

C. Revised preprocessing strategy

1. Separate the data into training and final test sets first, using a stratified split so the minority default class is represented in both sets. The final test set remains untouched until final evaluation.
2. On training data only, inspect missingness patterns. For income, prefer a robust or model-based/class-wise imputation strategy when justified; median imputation is often safer than the mean for highly skewed income. Add a missingness indicator if the fact that income is missing may itself carry information.
3. Retain occupation and handle its 18% missing values using a categorical most-frequent/class-wise/model-based approach, or create an explicit 'Unknown' category when substantively appropriate. Dropping the whole attribute is less attractive because 18% missing still leaves 82% potentially useful information.
4. Inspect income distribution. Because it is right-skewed, consider a log transformation for modelling and use quantile/equal-frequency discretization if discretization is genuinely required. Equal-width binning is rejected because it can create highly imbalanced bins.
5. Fit all scaling, imputation, transformation and discretization parameters using training data only. Apply the learned parameters unchanged to the test data.
6. Handle class imbalance using stratified sampling, class weights or an appropriate resampling strategy inside the training data. Do not oversample or otherwise manipulate the final test set.
7. Train a suitable classifier and select a decision threshold using validation data or cross-validation, prioritising defaulter recall subject to an acceptable precision/workload constraint.
8. Evaluate once on the untouched test set using confusion matrix, recall, precision, F1-score, balanced accuracy and accuracy. Compare the original and redesigned pipelines using exactly the same final test observations.

Why this design is preferred: It directly addresses leakage, skewness, missing data and class imbalance. Median/robust imputation is preferred to a global mean for highly skewed income because it is less affected by extreme values. Retaining occupation is preferred to dropping it because useful information remains available after appropriate missing-value handling. Stratified sampling/class weighting is preferred to an unstratified 10% random sample because the default class is rare.

D. Prediction before implementing

Aspect	Expected effect of redesign
Income dispersion	Median/robust imputation and a suitable transformation should preserve the distribution better than global-mean imputation; log transformation would reduce the influence of extreme high incomes.
Training class distribution	The training process should contain a more useful representation of the 3% default class through stratification and/or class weighting/resampling.
Accuracy vs defaulter recall	Accuracy may decrease or change only slightly, while recall for defaulters should improve substantially. Precision may decrease if the threshold is lowered, creating a deliberate trade-off.

E. Theory-based implementation and validation procedure

A suitable implementation can use Python with pandas and scikit-learn on a public credit dataset such as the UCI German Credit dataset or another documented credit-default dataset. The original and redesigned pipelines must use the same final test observations so that the comparison is fair.

Stage	Original pipeline	Redesigned pipeline
Split	After preprocessing; leakage risk.	Stratified train/test split first.
Missing income	Global mean of complete data.	Training-only median/robust/model-based imputation + optional missingness flag.
Occupation	Drop attribute.	Retain; impute/Unknown category on training data.
Income transformation	Equal-width 5 bins.	Log/robust transformation and, if required, training-only quantile bins.
Scaling	Complete-data min-max before split.	Fit scaler on training data only; transform test using training parameters.
Imbalance	Random 10% sample.	Stratified training and/or class weighting/resampling.
Evaluation	Accuracy emphasised.	Confusion matrix, recall, precision, F1, balanced accuracy and accuracy.

Because this document is intentionally theory-focused, no fabricated Python/Weka result is claimed here. The correct validation evidence for the submitted practical version is the actual tool output, settings, dataset link and screenshots requested by the assignment. The theoretical prediction is that the redesigned pipeline will sacrifice some majority-class accuracy if necessary to identify substantially more true defaulters.

F. Defence of the final design

The redesigned pipeline is preferable because the business cost of missing a real defaulter is much more important than obtaining a superficially high accuracy score. It prevents information leakage, treats missingness more carefully, preserves useful occupation information, respects the skewed income distribution and explicitly addresses the rare default class. The final decision threshold should be selected according to business cost rather than a default 0.5 threshold.

One remaining risk is dataset shift: the relationship between applicant characteristics and default may change over time because of economic conditions, lending policy or customer behaviour. Therefore, the model should be monitored after deployment and periodically revalidated on recent labelled data.

Overall Conclusion

The assignment demonstrates the complete data-mining workflow: selecting the appropriate mining functionality, following KDD, cleaning and integrating data, transforming and reducing data, discretising continuous attributes, and designing a leakage-safe predictive pipeline. The numerical sections show that data preparation and measure selection strongly influence the patterns obtained. In particular, outliers affect dispersion measures, poor imputation can distort distributions, unscaled attributes can dominate distance calculations, and class imbalance can make accuracy a misleading evaluation measure.